Rocks From the Moon's Far Side Help Solve a Mystery

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The side of the Moon we see from Earth looks very different from the side that always faces away. The near side has big, dark plains of old **volcanic** rock. The far side is rough, with lots of **craters** and a thicker **crust**.

In 2024, China's Chang'e-6 **mission** brought back the first rocks from the far side. A team led by Professor Hejiu Hui of Nanjing University studied them and compared them with near side rocks.

Deep inside, both sides seem very similar. But the outer layer, the **crust**, is different.

The scientists think they know why. Long ago, the Moon was covered by an ocean of melted rock. Later, the same side began to always face Earth. Earth was much closer then, and its **gravity** heated the near side more. This uneven heat stirred huge flows of melted rock and helped give the Moon two different faces.

Word Treasure

volcanic -- Volcanic describes rock or land made by volcanoes and lava.

craters -- Craters are bowl-shaped holes made when space rocks crash into a surface.

crust -- The crust is the hard outer layer of a planet or moon.

mission -- A mission is a planned trip to do a special job, like collecting rocks in space.

gravity -- Gravity is the force that pulls things toward each other, like a ball falling to the ground.

Background you need

The Moon always shows the same side to Earth, so we never see its far side from here.

Scientists study Moon rocks to learn how the Moon formed and changed.

China's Chang'e-6 was the first mission to bring back rocks from the far side.

Break it down

WHO: Professor Hejiu Hui's team at Nanjing University in China
WHAT: Studied the first rocks brought back from the Moon's far side
WHERE: The Moon, using samples from China's Chang'e-6 mission
WHY: To learn why the Moon's two sides look so different

Why it matters

Next time you look at the Moon, you will know its far side looks very different, and why.

Test what you remember

1. What covers much of the Moon's near side?

- (A) Big, dark plains of old volcanic rock
- (B) Forests
- (C) Oceans of water
- (D) Ice and snow

2. Which mission brought back the first rocks from the far side?

- (A) Mars Rover
- (B) Chang'e-6
- (C) Apollo 11
- (D) Voyager 1

3. In what year did the mission bring back the rocks?

- (A) 2030
- (B) 1969
- (C) 2000
- (D) 2024

4. Who led the team that studied the rocks?

- (A) A robot
- (B) An astronaut
- (C) Professor Hejiu Hui
- (D) A pilot

5. What part of the Moon is different on the two sides?

- (A) Its size
- (B) The crust
- (C) Its name
- (D) Its color at night

6. What heated the near side more long ago?

- (A) Earth's gravity
- (B) Fireworks
- (C) The Sun only
- (D) Rain

Answer key (try the quiz first!): 1: A 2: B 3: D 4: C 5: B 6: A

Questions to think about

MY THOUGHTS (at least 20 words):